



ROOT CAUSE UNLOCKED · CONDITION KIT

The Energy Root Cause *Kit*

Which tests actually explain fatigue, which ones do not, and exactly what to ask your doctor for

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The Energy Root Cause Kit

Which tests actually explain fatigue, which ones do not, and exactly what to ask your doctor for

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

See a doctor promptly, ahead of any of this, if your fatigue comes with unintentional weight loss, fever or night sweats, a new lump, blood in stool or urine, coughing blood, chest pain, breathlessness at rest or on mild exertion, fainting, new confusion, or severe headache. New fatigue in someone over 60, or fatigue that has worsened steadily over weeks rather than fluctuating, also deserves a medical workup first. Fatigue is a symptom of a great many things, and some of them are urgent.

If you have thoughts of harming yourself, please call or text 988 in the United States for the Suicide and Crisis Lifeline, or go to an emergency department.

Read this before you read anything else about exercise

If your symptoms reliably get **worse a day or two after exertion**, and that worsening is out of proportion to what you did, that is post-exertional malaise. It is the defining feature of myalgic encephalomyelitis and chronic fatigue syndrome, and it changes the entire plan.

Where post-exertional malaise is present, graded exercise is not the answer, and pushing through causes harm. Key 7 of *Unlock Your Energy* covers this and it is the most important safety line in the whole kit. Everything in these guides is written to be compatible with pacing rather than with pushing.

If that describes you, read the pacing section of the supplement guide before you change anything.

The eight keys, and what tests each one

Unlock Your Energy works through eight root causes. Blood work answers some of them cleanly, and for others the honest answer is that no panel will tell you.

Key	What it means	How you find it
1. Cellular	Mitochondrial energy production	No validated clinical test. Inferred, not measured
2. HPA axis	The stress response system and cortisol rhythm	Contested. See the honest section below
3. Sleep	Hours in bed are not hours of repair	Sleep study, and a two-week sleep log. Not a blood test
4. Gut	Absorption, and the microbial contribution	Celiac serology, stool analysis where symptoms justify it
5. Nutrients	Iron, B12, folate, magnesium, vitamin D	Blood work, and this is where the clearest answers live
6. Metabolic	Thyroid and blood sugar	TSH with free T4 and free T3, fasting glucose, HbA1c
7. Pacing	Post-exertional malaise and the energy envelope	History. The most consequential finding in this guide has no test
8. Testing	Choosing the right panel rather than the biggest one	This guide

Notice the shape of that. The two most decisive items, post-exertional malaise and sleep quality, are not blood tests. The nutrient key is the one where a lab draw genuinely changes what you do, which is why most of this guide is about it.

Key 5: nutrients, where the clearest answers are

Iron and ferritin, and the number that matters more than “normal”

This is the single highest-yield test in this guide, particularly if you menstruate.

The trial worth knowing. A double blind randomised placebo controlled trial in 144 non-anaemic women aged 18 to 55 with unexplained fatigue gave either oral iron or placebo. Iron improved fatigue, and the subgroup analysis is the part that matters: **only women with ferritin concentrations at or below 50 micrograms per litre improved** (PMID 12763985). A 2026 double-blind placebo-controlled randomised trial in non-anaemic iron-deficient women of reproductive age found iron supplementation reduced fatigue scores and raised ferritin (PMID 41736325).

Why this changes the conversation with your doctor. Most laboratory reference ranges call a ferritin of 15 or 20 normal. Those trials found benefit up to a threshold of 50. So “your iron is normal” and “your iron is high enough to not be causing your fatigue” are different statements, and the first does not establish the second.

What to ask for. Ferritin, serum iron, TIBC, transferrin saturation, and a complete blood count, together. Ferritin alone is not enough because it rises with inflammation and can look reassuring in someone genuinely depleted, which is why the hs-CRP below is worth having alongside it.

About that hs-CRP. It is a nonspecific screening marker. A raised result says an inflammatory load exists somewhere. It does not say where it is coming from, and it never confirms a diagnosis. Here it earns its place mostly as the sanity check on your ferritin.

The caution. Iron is not a supplement to take speculatively. Unnecessary iron causes constipation and nausea, overload is a real risk, and iron is a leading cause of poisoning death in young children. Test, treat with a real dose, and retest.

B12, folate, and homocysteine

Deficiency causes fatigue plus neurological symptoms that get attributed to almost anything else. Ask for B12, RBC folate, and homocysteine together, because homocysteine rises when either is functionally low and can reveal a problem that individual levels in the low-normal range hide.

A specific trap: long-term acid suppression and metformin both reduce B12 absorption. If you take either and nobody has checked, ask.

Vitamin D and magnesium

Worth testing, worth correcting a genuine shortfall, and worth honest expectations. Correcting a real deficiency is a legitimate part of a fatigue workup. Neither is a stimulant, and neither will fix fatigue that is being driven by unrefreshing sleep or an unrecognized pacing problem.

Red blood cell magnesium is generally the more informative of the two magnesium options, since most magnesium is intracellular.

Key 6: the metabolic thermostat

Thyroid

Ask for **TSH with free T4 and free T3**, not TSH alone. TSH by itself is the standard screen and it can miss a conversion problem that free T3 shows.

I have deliberately not published a functional target range for thyroid markers in this guide, because I only publish ranges I can transcribe exactly from a validated source and thyroid is one where the right target depends on age, pregnancy status, and whether you are already on replacement. That is a conversation with a clinician looking at your numbers, not a table.

If antibodies have never been checked and your thyroid is abnormal, TPO and thyroglobulin antibodies answer whether it is autoimmune, which changes the outlook. The companion Hashimoto's kit covers that in full.

Blood sugar

Fasting glucose, HbA1c, and fasting insulin. Fasting insulin is the one that gets left off and it is often the most informative of the three, because insulin can climb for years while fasting glucose still reads normal. The pattern to look for is not diabetes, it is the energy crash that follows a high-carbohydrate meal and the 3pm collapse that resolves with food. Those are worth catching early, and the diet guide covers what to do about them.

Key 2: the HPA axis, stated more honestly than usual

This is where fatigue marketing is at its worst, so here is the careful version.

What is not established. “Adrenal fatigue,” as a diagnosis in which chronic stress exhausts the adrenal glands into underproduction, is not a recognized clinical diagnosis, and no panel can confirm it because there is nothing validated to confirm.

What the actual research shows, which is more interesting. A 2026 meta-analysis examined the neuroendocrine signature of myalgic encephalomyelitis and chronic fatigue syndrome and found evidence for a bioactive cortisol deficit alongside exaggerated feedback sensitivity, integrating neuroendocrine and psychological perspectives to describe chronic stress maladaptation in fatigue syndromes (PMID 42026257).

Read the difference carefully, because it matters. That is not “your adrenals are tired.” It is a subtler regulatory finding in a specific diagnosed population, and it does not translate into a supplement that fixes it. What it does establish is that the stress axis is genuinely involved, which makes the load side of this a clinical target rather than a lifestyle suggestion.

Where cortisol testing does earn its place, and where it does not. I do use four-point diurnal cortisol testing, salivary, in my practice. What I use it for is the shape of the curve across a day, because a flat and low pattern and a wired and high pattern both present as exhaustion and they call for different approaches. Knowing which one you are looking at changes what I do next, and a single morning blood draw cannot show it.

Be clear about what that is and is not. It is a pattern I find clinically useful, not a validated diagnostic threshold, and it does not diagnose “adrenal fatigue” because that diagnosis does not exist to be made. Anyone selling you a saliva kit as proof of a named adrenal condition, or as the justification for a glandular

protocol, is going well beyond what the test can carry. Used for the pattern rather than for a label, it is worth running. Used as a diagnosis, it is not.

What I would actually do. If there is a clinical reason to suspect true adrenal insufficiency, which is a real and serious diagnosis, that is a morning cortisol and a physician. Otherwise, treat the stress axis through sleep, pacing, and load rather than through a panel.

Key 3 and Key 7: the two findings with no blood test

Sleep

Hours in bed are not hours of repair. Two things are worth ruling out before you conclude your fatigue is metabolic:

- **Sleep apnea.** If you snore, wake unrefreshed, wake with a headache, or someone has seen you stop breathing, ask for an assessment. Treating apnea treats the fatigue in a way no supplement will.
- **Insomnia as its own condition.** Cognitive behavioural therapy for insomnia has durable effects, with a systematic review and meta-analysis showing benefits on depressive symptoms sustained across the year following treatment (PMID 42435604), and fully automated digital versions have their own meta-analysis as well (PMID 42240717). The digital option matters because access is the usual barrier.

Bring a two-week sleep log rather than an impression. Bedtime, wake time, estimated hours, and a refreshed-or-not rating.

Post-exertional malaise

The highest-consequence finding in this guide, and it is answered by a question rather than a test: **do your symptoms reliably worsen a day or two after exertion?**

If yes, that reframes everything. The supplement guide covers pacing, and the principle is that you plan to your average day rather than your best one.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose.

Marker	Units	Functional target	Approximate standard range
Ferritin	ng/mL	50 to 150	25 to 300
Serum iron	ug/dL	65 to 110	40 to 150
TIBC	ug/dL	290 to 370	250 to 400
Transferrin saturation	%	20 to 40	15 to 45
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Homocysteine	umol/L	5 to 7	under 10.0
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Magnesium (serum)	mg/dL	2.0 to 2.5	1.5 to 2.7
Zinc (plasma)	ug/dL	70 to 110	60 to 120
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
Fasting insulin	uIU/mL	2.6 to 5.0	2.0 to 10.0
hs-CRP	mg/L	under 0.5	under 1.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0

The ferritin row is the one to argue about. The trial evidence above found benefit in women below 50, which is inside most labs' normal range.

On homocysteine, lower is not better. The 5 to 7 target is a window, not a floor. A homocysteine driven below it points toward overmethylation and B vitamin intolerance, which is its own problem rather than a better result. An elevated result is a prompt to find the short input, B12, folate, B6, thyroid, or kidney function, not a number to push down as far as it will go.

Reading your results as a pattern

Pattern one: the depleted menstruating woman. Ferritin under 50, heavy periods, fatigue worse in the second half of the cycle. This is the most common correctable pattern in this guide and it has the best trial support (PMID 12763985).

Pattern two: unrefreshing sleep. Everything normal, eight hours in bed, wake exhausted. The answer is a sleep assessment, not a bigger nutrient panel.

Pattern three: post-exertional malaise. Symptoms crash a day after exertion. Pacing is the intervention, and the risk here is a well-meaning exercise prescription making things worse.

Pattern four: everything normal and none of the above. Common, and not a dead end. It moves the work to load, sleep regularity, blood sugar stability, and mood, none of which show on a panel and all of which are addressable.

The conversation with your doctor

A request list you can bring

- CBC with differential
- Ferritin, serum iron, TIBC, transferrin saturation
- Vitamin B12, RBC folate, homocysteine
- 25-hydroxy vitamin D
- TSH with free T4 and free T3
- Fasting glucose, HbA1c, and fasting insulin
- hs-CRP, understood as a nonspecific screen rather than a diagnosis
- Celiac serology, drawn while still eating gluten
- A sleep apnea assessment if you snore or wake unrefreshed

The sentence that helps

“My ferritin is X. I have read that trials of iron for unexplained fatigue in non-anaemic women found benefit below 50, and I would like to discuss whether that applies to me.” That is specific, it is accurate, and it opens a real conversation instead of a reassurance.

What these labs cannot tell you

- **Whether your mitochondria are working.** There is no validated clinical test, and anyone selling one is ahead of the evidence.
- **Whether you have adrenal fatigue.** Not a recognized diagnosis, and a salivary panel does not establish it.
- **Whether you have post-exertional malaise.** That is your history, and it matters more than the panel.
- **Why you are tired, if everything is normal.** A clean panel is information, not dismissal.

When to seek clinical review

Promptly. Unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, new confusion. New or steadily worsening fatigue over 60.

Soon. Fatigue lasting more than six weeks without explanation, snoring or witnessed pauses in breathing, fatigue with low mood or loss of interest, or fatigue that reliably worsens a day after exertion.

Now: thoughts of harming yourself. Call or text 988.

Your next step

The companion guides do the rest: **The Energy Diet Guide** covers blood sugar and the gut, and **The Energy Supplement Guide** covers what has actually been studied, plus the pacing and sleep work that outperforms most of it.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

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If you would rather start on your own, start tracking. In fatigue, the pattern across weeks is the diagnosis:

Free symptom tracker: rootcauseunlocked.com/tracker

References

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Educational content only. Not medical advice, not a diagnosis, and not a substitute for your clinician. Persistent or unexplained fatigue deserves a medical workup.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

See a doctor before dieting your way through fatigue if it comes with unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, or new confusion, or if it began after age 60 or has worsened steadily over weeks.

If you are pregnant, nursing, underweight, diabetic, or have a history of disordered eating, talk with a clinician before restricting anything. In a fatigue population, an unnecessary restriction costs more than it does in most conditions, for the reason in the next section.

If you have thoughts of harming yourself, call or text 988.

The thing most fatigue diet advice gets wrong

Cooking is an activity. Shopping is an activity. Standing at a counter chopping vegetables for forty minutes is an activity, and it comes out of the same daily energy budget as everything else you need to do.

Almost every fatigue diet on the internet is written as though the reader has normal energy to spend on implementing it. Then it prescribes an elimination protocol with three home-cooked meals a day, and the person following it spends their entire envelope on food preparation and crashes.

So the first rule of this guide is that the plan has to cost less energy than it returns. Everything below is written to that constraint, and where a strategy is expensive to run, I say so.

If your symptoms reliably worsen a day or two after exertion, that is post-exertional malaise, and this constraint is not a nicety. Read the pacing section of the supplement guide first.

Where food sits among the eight keys

The eight root causes *Unlock Your Energy* works through are cellular, HPA axis, sleep, gut, nutrients, thyroid and blood sugar, pacing, and testing. The lab guide maps all eight and tells you which are yours.

Food is a direct lever on two of them, the nutrient key and the blood sugar half of the metabolic key, and an indirect one on the gut. It is not a lever on sleep apnea, on post-exertional malaise, or on a thyroid that needs replacing. Knowing which of your keys food can actually reach is what stops you spending a year on the wrong one.

Part one: blood sugar, the most reachable win

The 3pm collapse that resolves the moment you eat is the most fixable pattern in this guide, not a personality trait, and the fix is mechanical.

What to do:

- **Eat protein at breakfast.** The single change that most reliably flattens the rest of the day.
- **Do not go more than four or five hours without food** during your waking day.
- **Pair carbohydrate with protein, fat, or fiber** rather than eating it alone. A slice of toast alone and toast with eggs behave differently.
- **Watch liquid sugar specifically**, including the coffee drink that is functionally a dessert.
- **Notice what happens ninety minutes after you eat.** That is your data.

Sourcing note, stated plainly: the blood sugar guidance above is standard clinical nutrition and physiology rather than a finding from a trial in fatigue patients. I have not attached a citation to it because I did not find one that studied this pattern in this population. It is included because it is low risk, cheap, and reversible, and because the pattern is easy to test on yourself in a week.

Part two: eating for the iron finding

This section of the diet guide has actual trial evidence behind it, and it connects directly to the lab guide.

Why it matters. In a double blind randomised placebo controlled trial of 144 non-anaemic women with unexplained fatigue, iron improved fatigue, and the benefit was confined to women with **ferritin at or below 50** (PMID 12763985). A 2026 randomised placebo-controlled trial in non-anaemic iron-deficient women likewise found reduced fatigue and increased ferritin with supplementation (PMID 41736325).

What that means at the table. If your ferritin is low, diet supports repletion but rarely achieves it alone, so this section sits alongside the supplement conversation rather than replacing it.

Absorption is where the practical gains are.

- **Heme iron**, from red meat, poultry, and fish, absorbs far better than the non-heme iron in plants.
- **Vitamin C with the meal** improves non-heme absorption. Peppers, citrus, tomato, or strawberries alongside beans, lentils, or greens.
- **Tea and coffee block it.** Separate them from iron-rich meals and from any iron supplement by a couple of hours. This one change is free and it is the most common unforced error.
- **Calcium competes.** Do not take a calcium supplement at the same time as iron.
- **Cast iron cookware** contributes a small amount. That falls short of a treatment, though it still counts for something.

Part three: the gut key, without overclaiming it

The gut sits in Key 4 of the ebook, and it earns its place for two concrete reasons rather than a general terrain argument.

Absorption. You cannot correct an iron or B12 deficiency through a gut that is not absorbing, which is the practical link and the reason the lab guide asks for celiac serology.

Get celiac testing before you remove gluten. Celiac disease presents with fatigue often enough that it belongs in any fatigue workup, and the serology is only interpretable while you are still eating gluten. Remove it first and you lose the ability to get a clean answer without a deliberate gluten challenge later. This is the single most common way this diagnosis gets missed for years.

The bidirectional part. Sleep disruption and the gut interact, and a 2026 review of the sleep and gut-brain dialogue describes that bidirectional communication while being explicit that mechanistic and translational work is still needed (PMID 42092445). Directional rather than settled, and it points the same way as the sleep work in the supplement guide.

What I am not going to claim. That a gut protocol resolves fatigue. There is no trial supporting that, and in a population prone to spending scarce energy on complicated regimens, overclaiming here does real harm.

Part four: a plan that costs less energy than it returns

This is the section I would keep if I could only keep one.

Cook once, eat three times. Batch on your best day of the week, portion immediately, freeze. Not leftovers in the fridge for five days, actual frozen portions.

Keep a no-cook tier stocked at all times, for the days when cooking is not happening. Treat it as a normal part of the system rather than a failure plan. Stock tinned fish, pre-cooked rice, frozen vegetables, eggs, nut butter, plain yogurt, and fruit.

Lower the bar for what counts as a meal. Protein plus a vegetable plus a carbohydrate is a meal. It does not need to be assembled or attractive.

Use the shortcuts without guilt. Pre-chopped vegetables, frozen produce, rotisserie chicken, and grocery delivery all convert money into energy, which is the correct direction of trade in this condition.

Sit down to eat. Standing at the counter costs you more than you think, and digestion runs better when you are not rushing.

A sample rhythm rather than a menu. Batch two proteins and two vegetables on your best day. Cook a grain that keeps. Eat those in rotation for three days, then use the no-cook tier for two, then repeat. That is all the system asks.

What I deliberately left out, and why

- **Elimination diets, unless there is a specific reason.** They are expensive to run in energy terms, they narrow nutrition, and in an unexplained-fatigue population the yield is low. Celiac testing is the exception, and it is a blood test rather than a diet.
- **Extended fasting and aggressive intermittent fasting.** Skipping meals is the opposite of the blood sugar work above, and in a fatigue population it commonly backfires.
- **Ketogenic diets for fatigue.** Demanding to run, and I did not find trial evidence supporting them for unexplained fatigue.
- **“Energy” drinks, shots, and stimulant blends.** They borrow energy against later, and where post-exertional malaise is present, feeling able to do more than your envelope allows is actively harmful.
- **Detox and cleanse protocols.** No fatigue outcome evidence, real energy cost.
- **Green powders as a substitute for testing.** If your ferritin is 18, a scoop of powder is not the answer, and it will delay the real one.

Telling you what to skip is part of what you paid for, and in this condition it is also giving you your afternoon back.

Running a fair experiment

Weeks 1 and 2, baseline. Change nothing. Score energy out of ten daily and mark crash days.

Weeks 3 to 6, one change. Protein at breakfast is the one I would start with, because it is the cheapest to implement.

Week 7, judge it. Compare against baseline. If nothing moved, the lever is elsewhere, and the lab guide tells you where to look.

Then one more change. Never two at once.

Realistic expectations

What is realistic. A steadier afternoon within two weeks of fixing breakfast. Meaningfully better energy over two to three months if a real iron or B12 deficiency gets corrected.

What is not realistic. Diet resolving fatigue driven by untreated sleep apnea, post-exertional malaise, an unreplaced thyroid, or depression. Those need their own treatment.

And one thing worth saying plainly. If you have been told your fatigue is because you should eat better, and you have already been eating well for a year, the problem is not your willpower and it is not your diet. Go back to the lab guide.

When to seek professional care

Promptly. Unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, new confusion, or new fatigue over age 60.

Soon. Unexplained fatigue beyond six weeks, snoring or witnessed pauses in breathing, fatigue with low mood or loss of interest, a diet that has narrowed below roughly twenty foods, or a reliable crash a day after exertion.

Now. Thoughts of harming yourself. Call or text 988.

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your labs, your history, and whether post-exertional malaise is part of your picture.

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If you would rather start on your own, track first. In fatigue the pattern across weeks is the diagnosis:

Free symptom tracker: rootcauseunlocked.com/tracker

Read the companion documents alongside this one. **Understanding Your Energy Labs** covers the ferritin threshold worth arguing about, and **The Energy Supplement Guide** covers pacing and sleep, which outperform most of what you can buy.

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See a doctor before treating fatigue yourself if it comes with unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, or new confusion, or if it started after age 60 or has worsened steadily over weeks. Fatigue is a symptom of many things and some of them need finding.

If you have thoughts of harming yourself, call or text 988 in the United States, or go to an emergency department.

In the US, supplements are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed, with some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions. It names the funding and the evidence tier, and lets you see both.

The safety line that comes before everything else

If your symptoms reliably worsen a day or two after exertion, do not let anyone talk you into pushing through.

That pattern is post-exertional malaise, and it is the defining feature of myalgic encephalomyelitis and chronic fatigue syndrome. Where it is present, graded exercise is not the answer and can make you worse. That point is Key 7 of *Unlock Your Energy*, and it is the most important thing in this kit.

Everything below is written to work alongside pacing rather than against it.

Two foundations that beat everything you can buy

I would rather you spend your effort well than spend money here, so these come first.

Pacing, if post-exertional malaise is part of your picture

A scoping review of pacing, the energy-management approach used in ME/CFS, describes it as a prominent coping strategy while stating plainly that the literature base is not yet sufficient to dictate treatment practice (PMID 37838675). Co-production work with adults living with ME/CFS puts energy management at the center of daily life, alongside sleep, diet, and psychological support (PMID 39961545). Post-treatment and post-infectious fatigue syndromes are increasingly recognized as a group, with calls for better research and inclusive trial design (PMID 41967005).

The honest state of it: pacing is the consensus practical approach and the trial base behind it is thinner than it should be. It is also the approach least likely to harm you, which matters more than usual in a condition where the standard alternative caused harm.

What it looks like.

- Find your envelope on an **average** day, not your best one, and plan to the average.
- Break activity into short blocks with rest between, rather than one push followed by a crash.
- Stop before you are empty. The instinct to finish the task is the thing that costs you tomorrow.
- Expect a non-linear course and judge in months.

Sleep, treated as a treatment

Hours in bed are not hours of repair. Two things outrank every capsule below.

Rule out sleep apnea if you snore, wake unrefreshed, wake with headaches, or someone has seen you stop breathing.

And if the problem is insomnia, the treatment has real evidence. Cognitive behavioural therapy for insomnia produces effects on depressive symptoms sustained across the year after treatment in a systematic review and meta-analysis (PMID 42435604), and fully automated digital delivery has its own meta-analysis as well (PMID 42240717). The digital versions matter because access is usually the barrier, and they cost a fraction of a supplement stack.

How to read this guide

Test before you supplement, wherever a test exists. In fatigue this is not a formality, because the two nutrients with the best evidence only helped the people who were actually low.

Give it eight to twelve weeks, and track two numbers most days: energy out of ten, and whether you crashed after exertion.

One change at a time. Fatigue fluctuates on its own, and a stack teaches you nothing.

Tier 1: correct what is measurably low

Iron, only if your ferritin says so

The trial that defines this entry. In a double blind randomised placebo controlled trial of 144 non-anaemic women aged 18 to 55 with unexplained fatigue, oral iron improved fatigue, and the subgroup analysis found that **only women with ferritin at or below 50 micrograms per litre improved** (PMID 12763985). A 2026 double-blind placebo-controlled trial in non-anaemic iron-deficient women of reproductive age likewise found reduced fatigue scores and increased ferritin with iron supplementation (PMID 41736325).

How to read that. This is one of the cleanest findings in the whole fatigue literature, and it comes with a built-in filter: it worked in the depleted, and it did not work in everyone else. So the test comes first.

Typical studied range. 80 mg daily of elemental iron as ferrous sulphate in the 2003 trial (PMID 12763985). Dose and duration belong with your clinician, and a retest belongs at the end.

Cautions. Constipation and nausea are common and are the usual reason people quit. Iron overload is a real risk in people who do not need it. Separate iron from thyroid medication by four hours. **Keep it locked away from children**, since it is a leading cause of childhood poisoning death.

B12, folate, vitamin D, and magnesium, on the same principle

Correct a documented shortfall. Do not supplement speculatively and expect an energy effect, because that is not what the evidence supports. The lab guide covers what to ask for and what the functional targets are.

Two practical notes. Long-term acid suppression and metformin both reduce B12 absorption, so if you take either, ask for a level. And magnesium in the evening is well tolerated and may help sleep, which in this condition is a reasonable secondary reason to take it.

Tier 2: mitochondrial support, with honest expectations

Coenzyme Q10, with or without NADH

What the research shows. A prospective, randomized, double-blind, placebo-controlled trial tested CoQ10 plus NADH in people with myalgic encephalomyelitis and chronic fatigue syndrome, and reported benefit for perceived cognitive fatigue and health-related quality of life, describing the combination as generally safe while calling for further work to confirm the clinical benefits and clarify mechanism (PMID 34444817). A separate randomized, double-blinded, placebo-controlled study of ubiquinol-10 at 150 mg daily over 12 weeks in chronic fatigue syndrome found improvement in several symptoms, including measures of autonomic and cognitive function (PMID 27125909).

How to read it. Two small placebo-controlled trials in a diagnosed population, both positive, neither large. That is better than most things sold for fatigue and well short of settled. If you try it, try it for the right reason: modest improvement in perceived fatigue and cognition, not a return to your old energy.

Typical studied range. 150 mg daily of ubiquinol-10 for 12 weeks in the trial above (PMID 27125909). The combination trial used CoQ10 with NADH (PMID 34444817).

Timing. With a fat-containing meal, since absorption depends on it. Earlier in the day, as some find it activating.

Cautions. CoQ10 can reduce the effect of warfarin, which requires prescriber involvement. It may modestly lower blood pressure and blood sugar. Not established in pregnancy or nursing.

Tier 3: ask your prescriber first

- **Iron**, always, because it should follow a test and a dose decision rather than a guess.
- **Anything stimulant-like**, including high-dose caffeine products and “adrenal support” blends containing glandulars or licorice, particularly if you have high blood pressure. In a fatigue population these borrow energy rather than create it, and in post-exertional malaise they invite exactly the overexertion that causes the crash.
- **Thyroid glandulars and iodine**, which are not benign in autoimmune thyroid disease and are covered in the Hashimoto’s kit.
- **Anything at all in pregnancy or nursing.**

What I deliberately left out, and why

- **Adrenal glandular and adaptogen protocols sold on the back of an “adrenal fatigue” diagnosis.** That diagnosis is not recognized, and the real neuroendocrine finding in ME/CFS is subtler than the marketing, involving a bioactive cortisol deficit and exaggerated feedback sensitivity on meta-analysis (PMID 42026257). It does not translate into a product that fixes it. Note the distinction from the testing itself: I do use four-point diurnal cortisol clinically, for the shape of the curve rather than to name a condition, and the lab guide in this kit explains exactly what that does and does not tell you.
- **NAD precursors sold for fatigue.** Interesting biology, expensive, and the fatigue trial evidence does not yet justify the price. Note that the CoQ10 trial above used NADH, which is a different compound from the NMN and NR products being marketed.
- **Mitochondrial “support” stacks with a dozen ingredients at token doses.** You cannot tell what worked, and the doses that were studied are specific.
- **B12 injections in people whose B12 is normal.** Popular, and not supported.
- **Anything that promises energy quickly.** In a condition where overexertion causes crashes, a product that makes you feel able to do more than your envelope allows is not a good outcome.

Telling you what to skip is part of what you paid for.

Running a fair trial

Weeks 1 and 2, baseline. Change nothing. Score energy out of ten daily, and mark any day you crashed after exertion.

Weeks 3 to 14, one intervention. Start with whichever deficiency your labs actually showed. If nothing was low, start with the sleep work rather than a bottle.

Week 15, judge it honestly. Compare your last two weeks against baseline. In this condition, “fewer crashes” is often a better outcome signal than “more energy,” and it moves first.

Bring the tracker to appointments. Three months of daily scores is more useful than any description from memory.

Realistic expectations

What is realistic. Correcting a real iron or B12 deficiency can produce a genuine, noticeable change over two to three months. Everything else here is modest.

What is not realistic. A supplement resolving fatigue whose cause is unrefreshing sleep, an unrecognized pacing problem, untreated apnea, or depression. Those need their own treatment, and no amount of mitochondrial support substitutes.

Timeline. Eight to twelve weeks minimum. Iron often takes longer, because stores refill slowly.

Cost sense. If something has done nothing after twelve weeks at a sensible dose, stop paying for it.

When to seek professional care

Promptly. Unintentional weight loss, fever or night sweats, a new lump, bleeding, chest pain, breathlessness, fainting, new confusion, or new fatigue over age 60.

Soon. Unexplained fatigue beyond six weeks, snoring or witnessed apneas, fatigue with low mood or loss of interest, or a reliable crash a day after exertion.

Now. Thoughts of harming yourself. Call or text 988.

Where to find quality versions of these

Nothing here names a brand, on purpose. The evidence supports ingredients and forms.

Look for third-party testing. NSF, USP Verified, or equivalent.

Match the form to the trial. Ubiquinol-10 at 150 mg is what was studied, and it is not the same as a generic CoQ10 at an unstated dose.

Read the whole panel. Fatigue blends frequently hide caffeine or other stimulants, which is the last thing you want if crashes are your problem.

If you would rather not evaluate all of that yourself, I keep a screened dispensary through Fullscript, which ships directly to you. I hold no inventory myself, and everything in it is screened against the standards above.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the two highest-value items in this guide, pacing and sleep, cost nothing at all.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your labs, your history, and whether post-exertional malaise is part of your picture, because that single question changes the whole plan.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

Get in touch: rootcauseunlocked.com/contact

If you would rather start on your own, track first:

Free symptom tracker: rootcauseunlocked.com/tracker

Read the companion documents alongside this one. **Understanding Your Energy Labs** covers what to test and the ferritin threshold worth arguing about, and **The Energy Diet Guide** covers blood sugar and the gut.

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All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in [Energy-Kit-Citation-Ledger.md](#).

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